

ARTIFICIAL INTELLIGENCE

Aim

To solve a **Cryptarithmic Problem** using **Backtracking (Constraint Satisfaction)** in Python.

Algorithm

1. Start.
2. Take the words and the result (e.g., SEND + MORE = MONEY).
3. Identify all unique letters.
4. Assign a unique digit (0–9) to each letter.
5. Ensure no leading letter is assigned 0.
6. Check whether the arithmetic equation is satisfied.
7. If satisfied, display the solution.
8. Otherwise, try another digit assignment.
9. Stop.

Code:

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
digits = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

for perm in permutations(digits, len(letters)):
    d = dict(zip(letters, perm))

    # Leading digits cannot be zero
    if d['S'] == 0 or d['M'] == 0:
```

continue

```
send = (  
    d['S'] * 1000 +  
    d['E'] * 100 +  
    d['N'] * 10 +  
    d['D']  
)
```

```
more = (  
    d['M'] * 1000 +  
    d['O'] * 100 +  
    d['R'] * 10 +  
    d['E']  
)
```

```
money = (  
    d['M'] * 10000 +  
    d['O'] * 1000 +  
    d['N'] * 100 +  
    d['E'] * 10 +  
    d['Y']  
)
```

```
if send + more == money:  
    print("Solution Found")  
    print("SEND =", send)  
    print("MORE =", more)  
    print("MONEY =", money)
```

```
print("\nLetter Mapping:")

for k, v in d.items():

    print(k, "=", v)

break
```

OUTPUT:

```
==== RESTART: C:/Users/sri_jayu/OneDrive/Desktop/crypto.py =====
Solution Found
SEND = 9567
MORE = 1065
MONEY = 10652

Letter Mapping:
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
>>>
```

Ln: 51 Col: 0

Result:

Thus, the Cryptarithmic Problem ($SEND + MORE = MONEY$) was solved successfully using Constraint Satisfaction (Backtracking) in Python.